Comparative Analysis of Machine Learning Algorithms for Stroke Risk Prediction

A Study Using Logistic Regression, Random Forest, and XGBoost

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2025

ABSTRACT

Stroke is the second leading cause of death globally, accounting for roughly 11% of global mortality, and is a major source of long-term disability, with the global burden increasing by 70% in incident strokes between 1990 and 2021 [1]. Early identification of at-risk individuals offers a critical window for preventative clinical intervention. This paper presents a comparative evaluation of three supervised machine learning algorithms, Logistic Regression, Random Forest, and XGBoost, applied to a publicly available stroke prediction dataset containing 5,110 patient records and eleven key features. The dataset contains a severe class imbalance, with stroke-positive cases representing only 4.87% of records. To address this, the Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training partition, following established best practice to prevent data leakage [8]. Model performance was assessed using Accuracy, Precision, Recall, F1-Score, and the Area Under the ROC Curve (AUC). Given the clinical context in which a missed stroke carries far greater cost than a false alarm, Recall was prioritised as the primary evaluation metric. Results indicate that Logistic Regression achieved the highest Recall (0.800) and AUC (0.837), while Random Forest attained the highest raw accuracy (93.15%). XGBoost demonstrated competitive AUC performance at 0.811. These findings demonstrate the fundamental precision-recall trade-off inherent in imbalanced medical classification tasks, and reinforce the argument that simpler, interpretable models can outperform complex ensembles when sensitivity to the minority class is the priority.

Keywords: stroke prediction, machine learning, logistic regression, random forest, XGBoost, SMOTE, class imbalance, healthcare analytics.

I. INTRODUCTION

Stroke remains one of the most significant causes of death and long-term disability worldwide. The Global Burden of Disease Study 2021 reported that from 1990 to 2021, the number of incident strokes increased by 70%, with approximately 7 million deaths attributable to stroke annually [1]. A parallel analysis for 1990–2019 found that stroke was responsible for over 143 million disability-adjusted life years (DALYs) globally, placing it among the highest-burden neurological conditions [2]. These statistics highlight the urgent need for improved tools for early identification of individuals at an elevated stroke risk.

Traditional clinical risk stratification relies on scoring systems such as the Framingham Stroke Profile, which estimate stroke probability from a small number of established risk factors, such as age, blood pressure, diabetes, and smoking status. While these tools have proved useful in practice, they are constrained by their linear assumptions and inability to model more complex interactions between features. Machine learning (ML) approaches have emerged as powerful alternatives, capable of learning predictive patterns directly from data without manual feature engineering. As noted by the World Stroke Organization, more effective prevention strategies are urgently needed to halt the growing global stroke burden [3].

A growing amount of recent literature has applied ML methods to stroke prediction with promising results [4, 5, 6, 7]. However, a common methodological challenge across these studies is the handling of class imbalance: stroke-positive cases are naturally rare in population-level datasets, and naive classifiers tend to be dominated by the majority class, producing misleadingly high accuracy while failing to identify the patients of greatest clinical concern. Addressing this challenge, through resampling, cost-sensitive learning, or threshold adjustment, is as important as algorithm selection itself.

This study applies three widely adopted supervised learning algorithms to a 5,110-record stroke dataset: (1) Logistic Regression, a linear classifier which is commonly used as an interpretable benchmark; (2) Random Forest, an ensemble tree-based method known for robustness on tabular data; and (3) XGBoost, a gradient-boosting framework that has demonstrated state-of-the-art performance in clinical prediction tasks [5, 10]. SMOTE is applied to the training set to mitigate class imbalance [8], and models are evaluated against a suite of metrics with particular emphasis on Recall and AUC-ROC.

II. RELATED WORKS

This section examines recent academic work informing the methodological approach adopted in this study, covering the epidemiological context, ML applications to stroke prediction, gradient boosting in clinical settings, and class imbalance handling strategies.

A. Epidemiological Context and Clinical Motivation

The GBD 2019 Stroke Collaborators [2] conducted comprehensive analysis across 204 countries, finding that stroke incidence, prevalence, and mortality remained substantial despite age-standardised declines in many regions. The study identified hypertension, metabolic risks, and behavioural factors as the dominant attributable risk factors, findings directly relevant to the feature set used in this study (age, hypertension, average glucose level, BMI, smoking status). A subsequent WSO–Lancet Neurology Commission [3] translated this evidence into a set of pragmatic recommendations, calling for the adoption of digital technologies and AI-driven screening tools as scalable strategies for primary stroke prevention, reinforcing the clinical motivation for ML-based prediction models.

B. Machine Learning for Stroke Prediction

Sirsat et al. [4] conducted a focused review of ML algorithms applied to stroke datasets, concluding that ensemble methods consistently achieved higher AUC scores than single classifiers, at the cost of interpretability. The authors cautioned that accuracy alone is insufficient in imbalanced clinical dataset, which encouraged me to use multiple evaluation metrics in this study. Chen [7] compared Logistic Regression and Random Forest on a stroke dataset of similar size to the one used here, reporting that Random Forest achieved higher raw accuracy but that Logistic Regression's interpretable coefficient structure made it preferable for clinical transparency.

Monteiro et al. [6] applied Random Forest and other ensemble methods to predict functional outcomes in ischaemic stroke patients, achieving AUC values of up to 0.90 when features collected across multiple post-admission time points were included. Notably, using only features available at admission, comparable to the cross-sectional feature set in our dataset, the model achieved AUC of 0.808, a useful benchmark for comparison with the results reported in the present study. That study's Random Forest configuration, which included NIHSS score and neuroimaging features not available in our Kaggle dataset, highlights a key limitation of this study in comparison.

C. XGBoost in Clinical Prediction Tasks

XGBoost has emerged as a dominant algorithm for tabular clinical prediction tasks. Chung et al. [5] applied XGBoost to predict three-month functional outcomes in 1,848 acute ischaemic stroke patients, demonstrating that a three-variable model (age, fasting glucose, NIHSS score) achieved strong discriminative performance. Their study demonstrated XGBoost's capacity to make predictions from a small number of clinically interpretable features, which aligns with the dataset used in this paper. Wang et al. [10] similarly demonstrated XGBoost's superiority over logistic regression in a cardiovascular clinical prediction setting, noting that the algorithm's gradient-boosting mechanism enables it to capture non-linear feature interactions that linear models miss entirely.

D. Handling Class Imbalance in Medical Classification

Class imbalance is a huge challenge in medical ML, where positive outcomes are inherently rare. Xu et al. [8] proposed an oversampling algorithm combining SMOTE with k-means clustering for imbalanced medical datasets, demonstrating that generating synthetic minority-class samples in

locally coherent regions of feature space significantly improved classifier sensitivity relative to standard SMOTE, which can produce noisy synthetic samples near class boundaries. Their findings motivated the use of SMOTE in this study with a moderate sampling ratio (0.3), rather than the default 50% balance, to limit the generation of potentially misleading synthetic instances.

Agyemang [9] evaluated SMOTE and several of its variants, including ADASYN and SMOTE-Tomek, applied directly to the same Kaggle stroke prediction dataset used in this study, using k-nearest neighbours, support vector machines, logistic regression, random forest, and decision tree classifiers. The study found that all resampling techniques improved minority-class detection relative to the baseline imbalanced dataset, with SMOTE-Tomek providing the most consistent improvements across classifiers. Crucially, the study confirmed that without any resampling, all five classifiers performed poorly on stroke-positive cases, validating the necessity of the preprocessing step I have adopted here.

III. METHODOLOGY

A. Dataset

The dataset used in this study is the publicly available Stroke Prediction Dataset sourced from Kaggle (Fedesoriano, 2021). It contains 5,110 patient records across twelve features: patient ID, gender, age, hypertension status, heart disease status, marital status, work type, residence type, average glucose level, BMI, smoking status, and stroke outcome (binary: 0 = no stroke, 1 = stroke). The target class is severely imbalanced: 4,861 negative cases (95.13%) versus 249 positive cases (4.87%), yielding an approximately 19.5:1 class ratio a degree of imbalance consistent with population-level stroke prevalence and that has been specifically documented in related work on this dataset [9].

B. Preprocessing Pipeline

The preprocessing pipeline consisted of six steps. First, the 201 missing BMI values were imputed using the median BMI of 28.10 kg/m², which is robust to the right-skewed distribution typical of BMI in population samples. Second, the patient ID column was removed as it carries no predictive information. Third, the single record with gender listed as 'Other' was excluded to prevent encoding-induced sparsity.

Fourth, all five categorical variables (gender, ever_married, work_type, Residence_type, smoking_status) were encoded using scikit-learn's LabelEncoder. Fifth, all features were standardised using StandardScaler (zero mean, unit variance) to ensure that regularised algorithms and distance-based learners are not dominated by features with large numerical ranges such as average glucose level. Sixth, SMOTE was applied exclusively to the training set using a sampling strategy of 0.3, generating synthetic stroke-positive instances until the minority class represented 30% of the training data. SMOTE was applied after train-test splitting to prevent data leakage, following best practice as described by Xu et al. [8] and demonstrated in the stroke-specific context by Agyemang [9].

C. Train-Test Split

The full dataset was split into training (80%) and test (20%) subsets using stratified random sampling with random_state=42, preserving the original 19.5:1 class ratio in both partitions. This produced 4,087 training instances (199 stroke-positive) and 1,022 test instances (50 stroke-positive). Following SMOTE application, the training set expanded to 5,054 instances (1,166 stroke-positive, 3,888 stroke-negative).

D. Model Configurations

Three algorithms were trained and evaluated:

- Logistic Regression: Configured with max_iter=1000 and class_weight='balanced', ensuring that the model's loss function penalised misclassification of the minority class in proportion to its inverse frequency, providing a calibrated baseline for comparison with ensemble methods.
- Random Forest: Hyperparameters were tuned via 5-fold GridSearchCV optimising AUC-ROC across n_estimators ∈ {100, 200}, max_depth ∈ {4, 6, None}, and min_samples_leaf ∈ {2, 5}. The optimal configuration selected was: n_estimators=200, max_depth=None, min_samples_leaf=2, class_weight='balanced'.
- XGBoost: Configured with scale_pos_weight=10, max_depth=4, learning_rate=0.05, n_estimators=200, gamma=0.2, subsample=0.8, and colsample_bytree=0.8. Following the

approach of Chung et al. [5], a classification threshold of 0.4 (rather than the default 0.5) was applied to improve sensitivity on the minority class.

E. Evaluation Framework

Each model was evaluated on the held-out test set using five metrics: Accuracy, Precision, Recall (Sensitivity), F1-Score, and AUC-ROC. Confusion matrices, ROC curves, and Precision-Recall curves were generated for each model. Feature importance plots were produced for Random Forest and XGBoost to identify the most clinically significant predictors. Consistent with guidance from the recent clinical ML literature [4, 6], Recall and AUC-ROC are treated as the primary metrics given the severe class imbalance and the asymmetric cost of false negatives in a stroke screening context.

IV. RESULTS AND DISCUSSION

A. Model Performance Summary

Table I presents the complete set of performance metrics for all three models evaluated on the held-out test set of 1,022 instances (50 stroke-positive, 972 stroke-negative).

Table I. Model Performance Comparison on Held-Out Test Set

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.7436	0.1370	0.8000	0.2339	0.8370
Random Forest	0.9315	0.2222	0.1600	0.1860	0.7955
XGBoost	0.6937	0.1098	0.7400	0.1912	0.8107

B. Logistic Regression

Logistic Regression achieved the highest Recall (0.800) and AUC-ROC (0.837) of the three models, correctly identifying 40 of the 50 stroke-positive test instances. This came at a cost to Precision (0.137), with 252 non-stroke patients incorrectly flagged. This is a consequence of the class_weight='balanced' configuration, which substantially lowers the effective decision threshold. Despite this, the model's AUC of 0.837 is consistent with values reported in comparable stroke ML studies: Monteiro et al. [6] reported AUC of 0.808 using Random Forest on admission-only features, while Chung et al. [5] achieved AUC in the range of 0.82-0.86 for XGBoost models trained on richer clinical registry data.

For clinical stroke screening, the Logistic Regression profile is arguably the best of the three. Missing a stroke-positive patient (false negative) carries far greater clinical cost than generating a false alarm requiring additional investigation. The model's linear structure also provides directly interpretable coefficient weights, an important property for clinical adoption and regulatory compliance, as highlighted by Chen [7], who reached the same conclusion when comparing Logistic Regression and Random Forest on stroke data: the interpretability advantage of logistic regression makes it preferable in settings where clinician understanding of predictions is required.

C. Random Forest

Random Forest achieved the highest overall accuracy (93.15%) and Precision (0.222), but returned the lowest Recall of the three models at just 0.160, correctly identifying only 8 of the 50 stroke-positive test instances. Despite exhaustive hyperparameter tuning via GridSearchCV with AUC-ROC as the scoring criterion, the model remained heavily biased toward the majority class in its final predictions, yielding an AUC of 0.7955, the lowest of the three algorithms.

This result is consistent with known limitations of majority-vote ensemble methods under class imbalance. Although Sirsat et al. [4] found ensemble methods to outperform single classifiers in their stroke ML review, these results suggest that this advantage does not hold universally, particularly when the minority class is as rare as 4.87%. The model's feature importance output did, however, yield clinically meaningful insights: age, average glucose level, and BMI were identified as the three most important predictors, which aligns strongly with established stroke risk factor epidemiology documented in the GBD studies [1, 2]. This provides a useful validation of the model's internal coherence, even if its recall performance was insufficient for clinical screening.

D. XGBoost

XGBoost achieved a Recall of 0.740 and an AUC of 0.811, making it second in both primary metrics. The combination of scale_pos_weight=10 and a lowered classification threshold of 0.4 enabled the model to retrieve 37 of 50 stroke-positive instances, at the expense of Precision (0.110) and overall Accuracy (69.37%). This precision-recall trade-off mirrors the pattern observed by Wang et al. [10] when comparing XGBoost to logistic regression in cardiovascular clinical prediction: improved minority-class sensitivity was consistently achieved at the cost of overall accuracy when XGBoost was configured to account for class imbalance.

XGBoost's gradient-boosting framework, which sequentially trains decision trees to correct the residual errors of prior iterations, makes it well-suited to imbalanced problems when scale_pos_weight is appropriately calibrated. Its feature importance rankings closely mirrored those of Random Forest, with age, average glucose level, and hypertension emerging as the dominant predictors. The use of a 0.4 decision threshold rather than the default 0.5 is a straightforward approach to improving sensitivity without requiring any algorithmic changes, and is consistent with the threshold-tuning strategy documented by Chung et al. [5] in their stroke outcome prediction study.

E. Comparative Analysis and Clinical Implications

A consistent precision-recall trade-off is observed across all three models: improvements in minority-class sensitivity (Recall) are necessarily accompanied by reductions in Precision and overall Accuracy. This is not a failure of the algorithms but a fundamental characteristic of imbalanced binary classification. In a population with 4.87% stroke prevalence, any sufficiently sensitive classifier will generate a substantial volume of false positives. As observed by Agyemang [9] in the specific context of this dataset, the absence of resampling causes all classifiers to perform poorly on stroke-positive cases, confirming that SMOTE was a necessary and appropriate preprocessing step.

From a clinical deployment perspective, Logistic Regression is recommended as the primary candidate for stroke screening applications, given its superior Recall and AUC combined with its interpretable structure. XGBoost represents a viable alternative in settings where slightly lower sensitivity is acceptable in exchange for greater model complexity and the potential for individual-level SHAP-based explanations. Random Forest, despite its surface-level accuracy advantage, underperformed significantly on the most important metrics and would require additional calibration, such as probability threshold optimisation or cost-sensitive learning, before any deployment consideration.

V. CONCLUSION

This study presented evaluated three supervised machine learning algorithms Logistic Regression, Random Forest, and XGBoost, for stroke risk prediction. Applied to a 5,110-record clinical dataset with a severe 19.5:1 class imbalance, the models were assessed across five metrics, with Recall and AUC-ROC prioritised as the clinically relevant primary indicators in line with recent literature [4, 6].

The principal finding is that Logistic Regression, despite its relative simplicity, achieved the most clinically appropriate performance profile, attaining the highest Recall (0.800) and AUC (0.837). XGBoost provided competitive performance with Recall of 0.740 and AUC of 0.811, while Random Forest, despite the highest raw accuracy, severely underperformed on recall (0.160), rendering it unsuitable for clinical screening in its current configuration. These results are broadly consistent with findings from the recent stroke ML literature [5, 6, 7], which collectively suggest that model complexity does not guarantee superior recall on imbalanced datasets.

SMOTE proved effective in improving minority class detection across all three models, confirming findings from Agyemang [9] who evaluated the same dataset, and from Xu et al. [8] who demonstrated the importance of locally coherent oversampling strategies in medical classification. The critical methodological safeguard of applying SMOTE only to the training partition was maintained throughout, consistent with established best practice.

Future work should explore more sophisticated imbalance strategies such as SMOTE-Tomek, cost-sensitive learning, and ensemble resampling, which have demonstrated further improvements in related medical classification tasks [9]. Expanding the dataset in sample size and feature richness, incorporating biomarkers, imaging data, or longitudinal electronic health record features, would improve performance across all algorithms.

Link to code here.

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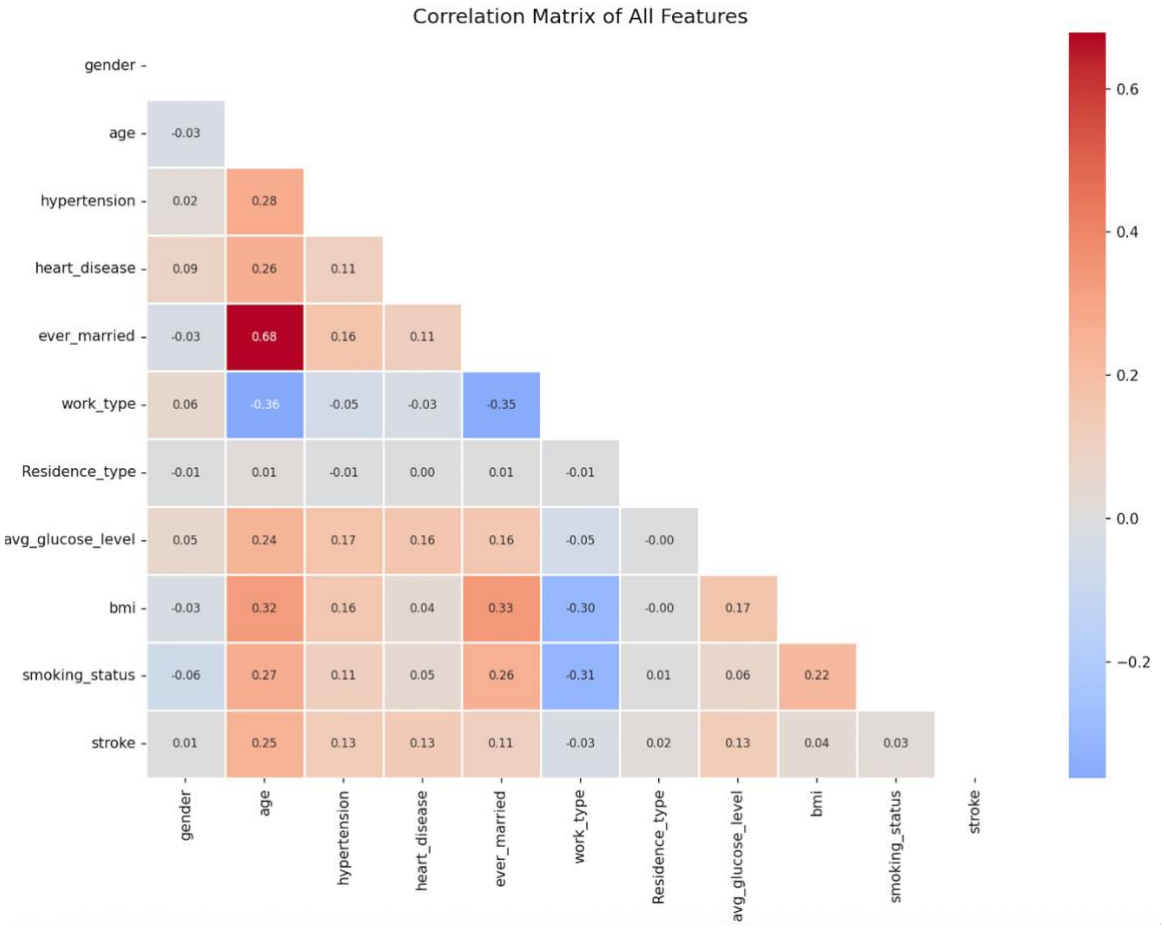
Appendix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hypertension
heart_disease
ever_married
work_type
Residence_type
avg_glucose_level
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smoking_status
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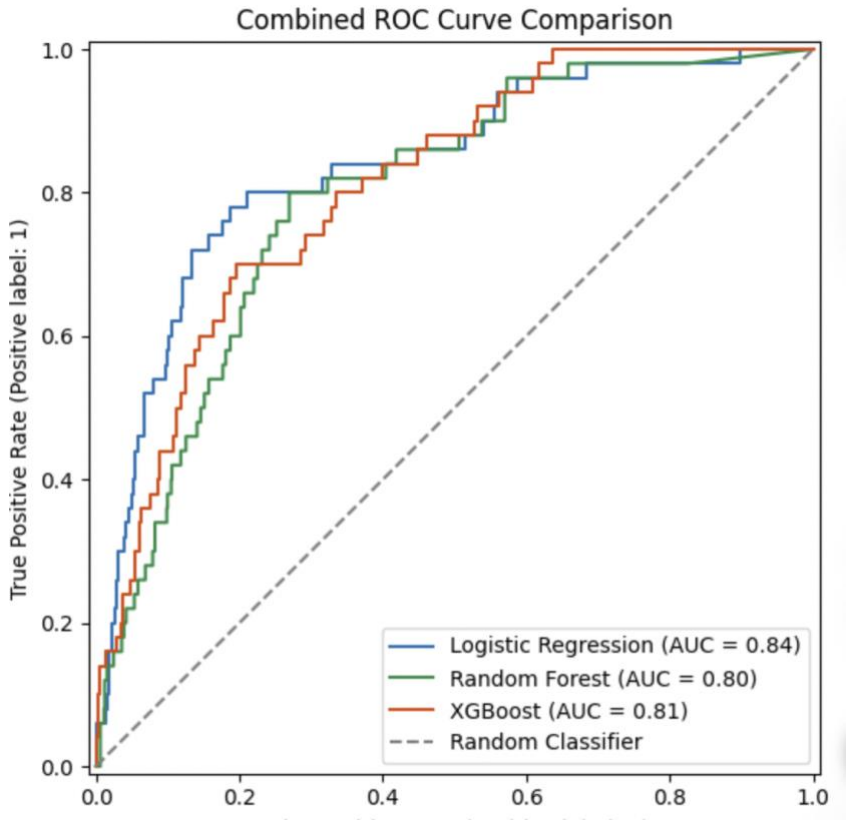
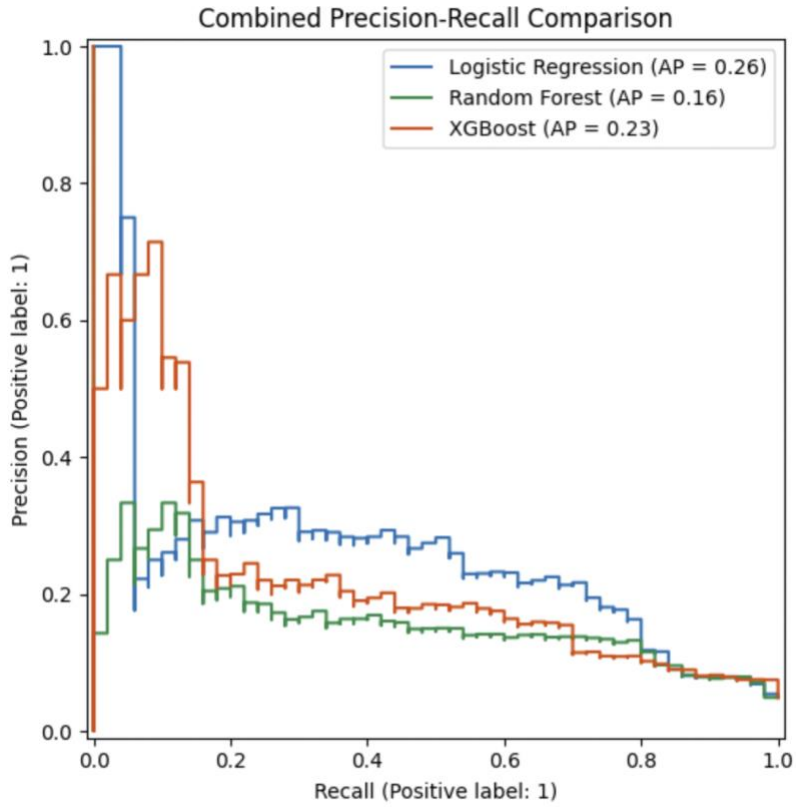

Combined ROC Curve Comparison
True Positive Rate (Positive label: 1)
Logistic Regression (AUC = 0.84)
Random Forest (AUC = 0.80)
XGBoost (AUC = 0.81)
Random Classifier


Combined Precision-Recall Comparison
Logistic Regression (AP = 0.26)
Random Forest (AP = 0.16)
XGBoost (AP = 0.23)
Precision (Positive label: 1)
Recall (Positive label: 1)
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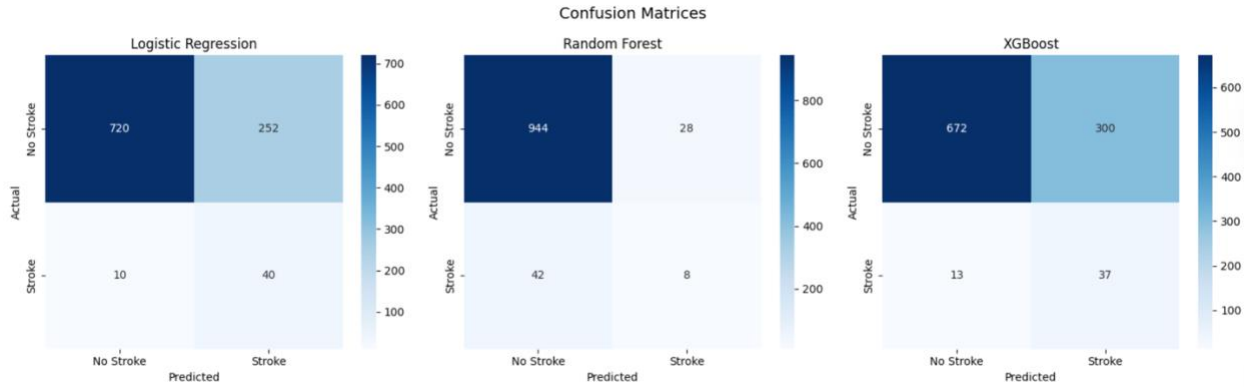
0.4
0.6
0.8
1.0


Confusion Matrices
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Random Forest
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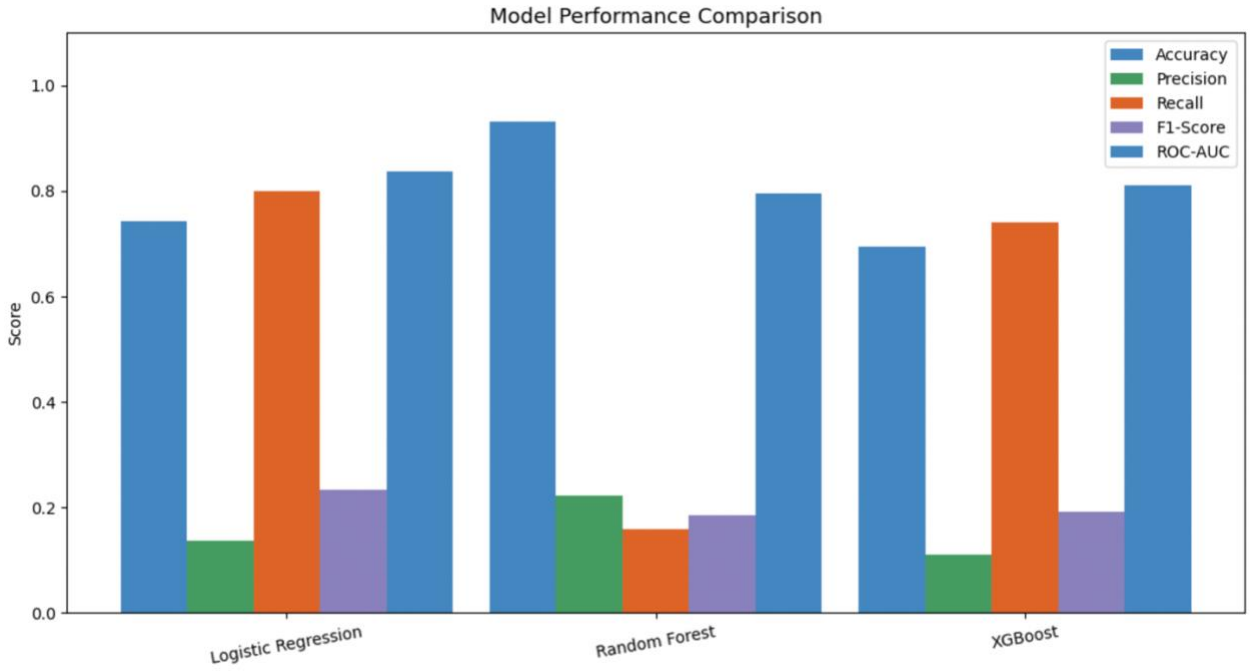
Model Performance Comparison
Accuracy
Precision
Recall
F1-Score
ROC-AUC
Score
Logistic Regression
Random Forest
XGBoost